BEAM Seminar Series

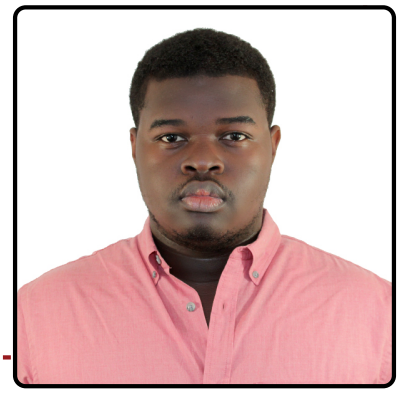
Engineering Mechanics Research Seminar

Saadbin Khan
FINLAB

Influence of tracheal microstructure on insect respiratory flows

Yogesh Phalak
Pendar Group

How Snakes Regulate Weight Distribution to Compensate for the Lack of Frictional Anisotropy



Kesna Fairclough
Comp. Mech. Lab

Torsion of Residually Stressed Mooney-Rivlin Cylinders

Time & Place:
Wednesday, December 6th
2:30 - 3:45 pm
Surge Building, Room 104A

COLLEGE OF ENGINEERING
BIOMEDICAL ENGINEERING AND MECHANICS
VIRGINIA TECH

BEAM Seminar Series

Saadbin Khan
FINLAB

Influence of tracheal microstructure on insect respiratory flows

Abstract:

Insects breathe using a network of tracheal tubes that bring respiratory gases directly to the tissues without using an intermediate oxygen carrier like blood. This mechanism inspires advancements in microfluidic technology; however, our understanding of the fundamental principles of insect respiration is limited. In this study, we consider the effect of irregularities in the tracheal wall, which contains local, periodic, 'bump'-like, microscale thickenings called taenidia. We carried out 3D finite-volume simulations of air flow in idealized tracheal models and analyzed the effects of the geometric parameters of the taenidia. We found that the presence of taenidia can accelerate peak flow, enhancing longitudinal dispersion. We further compared the flow in the idealized trachea to the flow through an anatomically accurate tracheal model from the darkling beetle Zophobas morio and found that small differences from the idealized geometries could lead to significant differences in flow patterns.

Bio:

Saadbin Khan is a third-year PhD student in the Engineering Mechanics program at Virginia Tech. He works with Dr. Anne Staples in the FINLAB, exploring and modeling the intricate dynamics of how insects breathe. Additionally, he is a BIOTRANS student and serves as a graduate administrative assistant for the program. Saadbin earned his BS in Mechanical Engineering from Bangladesh University of Engineering and Technology and an MS in Mechanical and Aerospace Engineering from Oklahoma State University. Currently, he is working on an NSF project focusing on the role of hydrodynamic slip and fine tracheal geometry in insect respiratory flow.

Time & Place:

Wednesday, December 6th
2:30 - 2:55 pm
Surge Building, Room 104A

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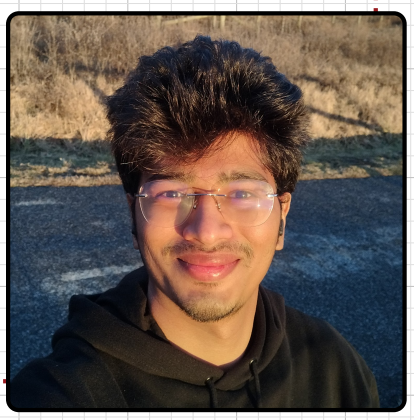

Yogesh Phalak
Pendar Group

How Snakes Regulate Weight Distribution to Compensate for the Lack of Frictional Anisotropy

Abstract:

Snake locomotion, a captivating biomechanical study, heavily relies on frictional anisotropy for generating diverse locomotion patterns. Studies have highlighted the significance of this anisotropy, particularly in facilitating snake gaits across varied terrains. However, encountering surfaces devoid of frictional anisotropy, such as smooth surfaces or granular media, poses a challenge. This research investigates how snakes dynamically adjust weight distribution to compensate for the absence of frictional anisotropy. Drawing inspiration from the resistive force theory framework for microbial locomotion, we propose a parallel mathematical-robophysical modeling approach. This approach aims to capture the intricate compensatory mechanisms employed by snakes. We employ a snake-like robot equipped with actuators for shape control and normal force distribution to validate our models. Ongoing work focuses on optimizing biomechanical aspects, bridging the gap between simulations and real-world snake biomechanics, and further exploring microbial-inspired compensatory strategies.

Bio:

Yogesh Phalak is a second-year Ph.D. student in engineering mechanics at Virginia Tech, concurrently working on a master's under the guidance of Dr. Hodjat Pendar. With a bachelor's degree in Electrical and Electronics Engineering from Visvesvaraya National Institute of Technology (2020), Yogesh's primary research focus for this master's degree is on the biomechanical study and optimization of snake locomotion gaits and their robophysical modeling. Additionally, Yogesh collaborates with Dr. Suyi Li at Dynamic Matter Lab, primarily working on Physical reservoir computing and Reduced order modeling on Origami and Kirigami structures and soft robotics

Time & Place:

Wednesday, December 6th
2:55 - 3:20 pm
Surge Building, Room 104A

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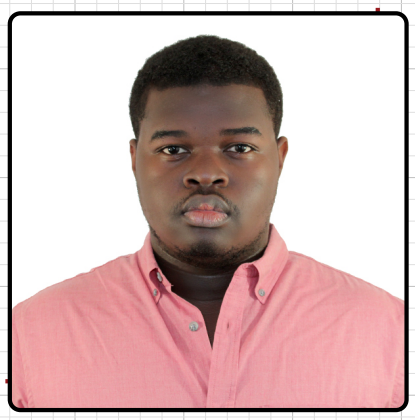

Kesna Fairclough
Comp. Mech. Lab

Torsion of Residually Stressed Mooney-Rivlin Cylinders

Abstract:

This presentation explores the torsional behavior of functionally graded, residually stressed Mooney-Rivlin materials with residual stresses induced from a Volterra dislocation. Functionally graded materials (FGMs) are composed of two or more constituents, with volume fractions and moduli continuously varying in at least one spatial direction. The study provides closed-form solutions that relate the axial stretch and the twist per unit length to the axial force, the resultant torque, and the through-thickness stresses. We also utilize the Polynomial Search-Differential Evolution algorithm to determine the variation in material moduli required to achieve a specified stress distribution.

Bio:

Kesna Fairclough is a dedicated PhD student who joined Virginia Tech in 2021 and has been working in the Batra Computational Mechanics Lab. His research interests include Continuum Modelling, Robust Optimization, and the Physics of Crystals. Kesna earned his undergraduate degree in Applied Mathematics and Materials Science (specialization in Crystallography) from the University of the West Indies in 2020. Since then, he has been passionately pursuing advanced studies and striving to make significant contributions to the field of computational mechanics.

Time & Place:

Wednesday, December 6th
3:20 - 3:45 pm
Surge Building, Room 104A

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